

CSA07-CN_CO3-AT3

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A university connects its Engineering, Medical, and Arts campuses—each running ^{*} a different LAN technology—into one seamless network using routers that translate between them. What is this combined network called?

- ☐ A single broadcast domain
- ☒ An internetwork
- ☐ A virtual LAN (VLAN)
- ☐ A collision domain

Feedback

Correct Answer! Well done.

An ISP wants to guarantee a fixed, dedicated bandwidth path for a video call for ^{*} its entire duration, even if no data is being sent momentarily. Which switching technique should it use?

- ☐ Packet switching
- ☐ Datagram switching
- ☒ Circuit switching
- ☐ Message switching

Feedback

Correct Answer! Well done.



During a live cricket stream, network congestion causes some packets to arrive out of order at the viewer's device, yet the video still plays smoothly after buffering. Which switching approach explains why packets could take different paths and arrive out of order? *

- ☐ Circuit switching
- ☒ Datagram packet switching
- ☐ Virtual-circuit switching
- ☐ Time-division multiplexing

Feedback

Correct Answer! Well done.

A network administrator notices that even though two packets belong to the same file transfer, they took different physical paths through the network and arrived at slightly different times. Which core property of the datagram approach explains this? *

- ☒ Each packet is routed independently based on current network conditions
- ☐ All packets follow a pre-established fixed path
- ☐ Packets are converted to circuits before transmission
- ☐ The network guarantees in-order delivery at Layer 3

Feedback

Correct Answer! Well done.



A company wants efficient use of shared links so that bandwidth is not wasted when a host is idle, unlike a dedicated telephone line. Which technique best supports this efficient statistical sharing of link capacity? *

- ☐ Circuit switching
- ☒ Frequency Division Multiplexing only
- ☐ Packet switching
- ☐ Dedicated leased line

Feedback

Incorrect Answer. Please review and try again!

In a datagram network, Router R2 crashes for 10 seconds. Packets that were being routed through R2 for an ongoing file download are rerouted through R3 without disrupting the whole connection. What key advantage of datagram (connectionless) networks does this scenario illustrate? *

- ☐ Guaranteed fixed delay for every packet
- ☒ Resilience to node failure through independent routing
- ☐ Reduced header overhead
- ☐ Mandatory reservation of resources before transmission

Feedback

Correct Answer! Well done.



A network engineer is designing WAN links between two data centers that exchange bursty, unpredictable traffic (sometimes idle, sometimes a huge spike). Why would packet switching be more cost-effective here than circuit switching? *

- ☒ Because circuit switching reserves bandwidth even during idle periods, wasting capacity
- ☐ Because packet switching requires dedicated end-to-end circuits
- ☐ Because packet switching guarantees zero delay
- ☐ Because circuit switching cannot handle bursty traffic at all

Feedback

Correct Answer! Well done.

An internetwork consists of Network A (Ethernet) and Network B (Wi-Fi) joined by a router. A host on Network A wants to send data to a host on Network B. What is the minimum functionality the router must provide for this internetwork to function? *

- ☐ Convert the physical medium only, without touching addressing
- ☒ Forward packets between networks using Layer 3 (IP) addressing and route selection
- ☐ Simply repeat electrical signals between the two networks
- ☐ Translate MAC addresses into IP addresses only, with no forwarding logic

Feedback

Correct Answer! Well done.



A telecom operator sets up a fixed path through several switches for a phone call; * all voice data uses that same path until the call ends, after which the path is released. Later analysis shows the line was idle 40% of the call but still occupied. Which switching method's known drawback does this best illustrate?

- ☐ Packet switching's store-and-forward delay
- ☒ Circuit switching's inefficient use of reserved bandwidth during idle periods
- ☐ Datagram approach's lack of a fixed path
- ☐ Message switching's high storage requirement

Feedback

Correct Answer! Well done.

A researcher argues that the Internet works because it is 'a network of networks' * rather than a single homogeneous network. Which concept does this statement directly describe?

- ☐ Subnetting
- ☒ Internetworking
- ☐ Multicasting
- ☐ Circuit switching

Feedback

Correct Answer! Well done.



In a packet-switched datagram network, each packet carries a full destination address and is treated independently by routers, with no connection setup phase. What is the primary trade-off of this design compared to virtual-circuit (connection-oriented) packet switching? *

- ☒ Higher per-packet header overhead and no guarantee of in-order delivery, but greater flexibility and fault tolerance
- ☐ Lower reliability because packets cannot be routed around failures
- ☐ Fixed bandwidth reservation that wastes capacity
- ☐ Mandatory call setup before every packet is sent

Feedback

Correct Answer! Well done.

A store-and-forward switch receives an entire packet, checks it for errors, and only then forwards it to the next hop, rather than starting to forward bits before the whole packet arrives. What is the main consequence of this behavior for a chain of several switches? *

- ☐ It eliminates all forwarding delay
- ☒ It introduces cumulative delay proportional to the number of hops and packet size
- ☐ It guarantees packets arrive in order
- ☐ It prevents the need for routing tables

Feedback

Correct Answer! Well done.



A financial firm needs predictable, constant delay for a real-time trading feed and * is willing to pay for reserved bandwidth even when idle. A cloud provider needs to serve millions of small, bursty web requests cheaply. Which switching techniques best match the firm's and the provider's needs respectively?

- ☐ Both should use packet switching
- ☒ Firm: circuit switching; Provider: packet switching
- ☐ Firm: packet switching; Provider: circuit switching
- ☐ Both should use message switching

Feedback

Correct Answer! Well done.

A packet in a datagram network is dropped by an intermediate router because its * buffer is full. The source is never notified directly by the network layer. Which characteristic of datagram (connectionless) networks does this reflect?

- ☐ The network layer provides guaranteed reliable delivery
- ☒ Best-effort delivery, where reliability must be handled by a higher layer if needed
- ☐ Circuit reservation prevents any packet loss
- ☐ Every packet loss triggers automatic network-wide rerouting

Feedback

Correct Answer! Well done.



Two companies merge and must connect their previously separate networks, which use different addressing schemes and media types, into a unified system that still allows any host to reach any other host. Which networking concept is central to solving this problem? *

- ☐ Multicasting
- ☒ Internetworking via routers performing Layer 3 forwarding
- ☐ Circuit switching between LAN segments
- ☐ Increasing the MTU of all devices

Feedback

Correct Answer! Well done.

A company is assigned the block [192.168.10.0/24](#) and must create 4 equally sized subnets for 4 departments. What is the new subnet mask and how many usable hosts does each subnet support? *

- ☐ /26, 62 usable hosts
- ☒ /25, 126 usable hosts
- ☐ /27, 30 usable hosts
- ☐ /28, 14 usable hosts

Feedback

Incorrect Answer. Please review and try again!



A network administrator needs subnets for departments with 100, 50, 25, and 10 * hosts from [172.16.0.0/16](#) using VLSM to minimize wasted addresses. What is the main advantage of VLSM over fixed-length subnetting in this scenario?

- ☐ It forces all subnets to be the same size for simplicity
- ☒ It allows subnet sizes to match actual host requirements, reducing address waste
- ☐ It eliminates the need for a subnet mask entirely
- ☐ It converts the network to IPv6 automatically

Feedback

Correct Answer! Well done.

Given the IP address 10.20.30.40 with subnet mask 255.255.255.192, which * subnet does this host belong to, and what is the broadcast address of that subnet?

- ☐ Network 10.20.30.0, broadcast 10.20.30.63
- ☒ Network 10.20.30.32, broadcast 10.20.30.63
- ☐ Network 10.20.30.0, broadcast 10.20.30.255
- ☐ Network 10.20.30.64, broadcast 10.20.30.127

Feedback

Correct Answer! Well done.



A host is configured with IP [192.168.5.130/26](#). Without a working DHCP server, the admin must manually determine its default gateway from a list of candidate addresses. Which of these could be a valid gateway on the same subnet as this host? *

- ☐ 192.168.5.65
- ☐ 192.168.5.129
- ☒ 192.168.5.190
- ☐ 192.168.5.193

Feedback

Incorrect Answer. Please review and try again!

An organization has the address block [200.10.5.0/24](#) and wants to create the maximum number of subnets while still supporting at least 20 hosts per subnet. What subnet mask should be used? *

- ☐ /25
- ☒ /26
- ☐ /27
- ☐ /28

Feedback

Incorrect Answer. Please review and try again!



In a Distance Vector routing protocol like RIP, Router A advertises a route to Network X with a hop count of 2. Router B, directly connected to A, then advertises this route onward with hop count 3. Suddenly Network X goes down, but due to delayed updates, Router B still believes it can reach X via A at increasing hop counts. What classic problem does this describe? *

- ☐ Split horizon violation
- ☒ Count-to-infinity problem
- ☐ Link-state flooding storm
- ☐ Dijkstra convergence failure

Feedback

Correct Answer! Well done.

To prevent the count-to-infinity problem in distance vector routing, a protocol is designed so that a router never advertises a route back to the neighbor from which it learned that route. What technique is this? *

- ☐ Poison reverse
- ☒ Split horizon
- ☐ Hold-down timer
- ☐ Triggered update

Feedback

Correct Answer! Well done.



A network uses Link State routing (like OSPF). When a link fails, every router floods a Link State Advertisement (LSA) describing the change to all other routers in the area, and each router independently recomputes shortest paths. Compared to Distance Vector, why does this design generally converge faster and avoid routing loops? *

- ☐ Because routers only share information with their immediate neighbors, exactly like Distance Vector
- ☒ Because every router builds a complete, identical topology map and runs Dijkstra's algorithm independently, rather than relying on second-hand hop-count rumors
- ☐ Because Link State routing does not need to recompute routes after failures
- ☐ Because Link State routing uses circuit switching internally

Feedback

Correct Answer! Well done.

A live video conferencing platform wants to send the same stream to 500 participants across different networks without sending 500 separate copies from the source. Which IP addressing/delivery technique is designed exactly for this? *

- ☐ Unicast addressing
- ☐ Broadcast addressing
- ☒ Multicast addressing (e.g., Class D / 224.0.0.0-239.255.255.255)
- ☐ Anycast addressing

Feedback

Correct Answer! Well done.



A network technician runs a diagnostic tool that sends a packet with an intentionally low Time-To-Live (TTL) to each hop along the path to a destination, receiving 'TTL exceeded' messages from each router in turn to map the route. Which protocol underlies this tool (e.g., traceroute)? *

- ☐ ARP
- ☒ ICMP
- ☐ IGMP
- ☐ BGP

Feedback

Correct Answer! Well done.

A user reports 'Destination Unreachable' errors when trying to reach a server. The network layer itself generates these error-reporting messages back to the source rather than silently dropping packets. Which protocol is responsible for generating such error messages? *

- ☒ TCP
- ☐ ICMP
- ☐ UDP
- ☐ DNS

Feedback

Incorrect Answer. Please review and try again!



An organization is running out of public IPv4 addresses and also struggles with complex header processing at high-speed routers. A new engineer proposes migrating to IPv6. Which of the following is a genuine structural advantage of IPv6 addressing in this scenario? *

- ☐ IPv6 addresses are shorter than IPv4, saving header space
- ☒ IPv6 provides a vastly larger 128-bit address space and a simplified, fixed-length header that speeds up router processing
- ☐ IPv6 requires NAT for every internal host, improving security
- ☐ IPv6 uses the same 32-bit addressing as IPv4 but with more classes

Feedback

Correct Answer! Well done.

A campus network is subnetted as [172.20.16.0/20](#). A new lab needs to know the range of usable host addresses within this subnet. What is the correct usable host range? *

- ☒ 172.20.16.1 to 172.20.31.254
- ☐ 172.20.16.0 to 172.20.31.255
- ☐ 172.20.16.1 to 172.20.16.254
- ☐ 172.20.0.1 to 172.20.255.254

Feedback

Correct Answer! Well done.



A company's router table has two routes to the same destination network: one learned via RIP with hop count 4, and one learned via OSPF with a lower administrative distance. The router chooses the OSPF route. What routing principle is being applied? *

- ☐ Longest prefix match
- ☒ Administrative distance determines trust between different routing protocols when multiple sources advertise the same destination
- ☐ Split horizon
- ☐ Poison reverse

Feedback

Correct Answer! Well done.

A router receives a packet destined for 192.168.4.55 and has two matching routing table entries: [192.168.4.0/24](#) and [192.168.4.0/26](#). Which entry will the router use to forward the packet, and why? *

- ☐ [192.168.4.0/24](#), because shorter prefixes are always preferred
- ☒ [192.168.4.0/26](#), because routers use the longest prefix match (most specific route) rule
- ☐ Both equally, using load balancing by default
- ☐ Neither; the packet is dropped due to ambiguity

Feedback

Correct Answer! Well done.



A small office is assigned the private address block [192.168.1.0/28](#) for a point-to-point link between two routers, which only ever needs 2 usable host addresses. Why might the network engineer instead choose a /30 subnet for this specific link? *

- ☒ A /30 provides exactly 2 usable addresses, minimizing wasted IP space for a link needing only 2 hosts
- ☐ A /30 supports multicast better than /28
- ☐ A /30 is required by IPv6 standards
- ☐ A /28 cannot be used for point-to-point links at all

Feedback

Correct Answer! Well done.

In a distance vector routing network, Router A, B, and C form a triangle. Link A-B fails. Router A still hears from Router C that C can reach B in 2 hops (via A), creating a routing loop where A and C keep incrementing hop counts for the unreachable network. Which mechanism, if properly implemented, would have prevented this loop from forming in the first place? *

- ☐ Increasing the MTU of the links
- ☒ Split horizon with poison reverse
- ☐ Enabling multicast routing
- ☐ Switching to IPv6 addressing

Feedback

Correct Answer! Well done.



An OSPF network experiences a link cost change. Within seconds, all routers in the area have updated, identical views of the topology and recompute optimal paths using Dijkstra's algorithm. Why is this approach called 'link state' rather than 'distance vector'? *

- ☐ Because it only shares the total distance to each destination with neighbors
- ☒ Because each router shares information about the state of its own links with the entire area, not just estimated distances with neighbors
- ☐ Because it uses circuit switching for updates
- ☐ Because it avoids using any routing algorithm

Feedback

Correct Answer! Well done.

A network needs to send a video stream to a specific group of subscribed routers/hosts (like an IPTV service) rather than to everyone (broadcast) or just one host (unicast). Which IPv4 address range and protocol combination supports dynamic group membership for this purpose? *

- ☐ Class A addresses with ARP
- ☒ Class D addresses (224.0.0.0-239.255.255.255) with IGMP for group membership management
- ☐ Class B addresses with ICMP
- ☐ Loopback addresses with RIP

Feedback

Correct Answer! Well done.



A student calculates that the block [10.0.0.0/8](#) can be subnetted using a /12 mask. *
How many subnets does this create, and how many hosts does each subnet support?

- ☒ 16 subnets, 1,048,574 hosts each
- ☐ 256 subnets, 65,534 hosts each
- ☐ 4096 subnets, 4094 hosts each
- ☐ 16 subnets, 65,534 hosts each

Feedback

Correct Answer! Well done.

A network administrator wants to summarize (aggregate) the routes [192.168.16.0/24](#) through [192.168.19.0/24](#) into a single routing table entry to reduce table size. What is the correct summarized route? *

- ☒ [192.168.16.0/22](#)
- ☐ [192.168.16.0/23](#)
- ☐ [192.168.0.0/16](#)
- ☐ [192.168.16.0/25](#)

Feedback

Correct Answer! Well done.



A router pings a neighboring router and receives an ICMP 'Time Exceeded' message even though the destination is reachable and only 3 hops away, because the packet's TTL was mistakenly set to 1. What does this reveal about how TTL and ICMP interact in IP networks? *

- ☐ TTL has no effect on ICMP behavior
- ☒ Each router decrements TTL by 1; when TTL reaches 0, the router discards the packet and sends an ICMP Time Exceeded message back to the source
- ☐ ICMP messages are only generated by the destination host, never intermediate routers
- ☐ TTL is only relevant in IPv6, not IPv4

Feedback

Correct Answer! Well done.

A company's link-state routing protocol experiences a scenario where two routers momentarily disagree about network topology right after a link goes down, but within a few seconds all routers converge to a consistent view using flooded LSAs and sequence numbers. What does this convergence behavior best demonstrate? *

- ☐ Link-state protocols never experience temporary inconsistency
- ☒ Link-state protocols use reliable flooding with sequence/aging numbers so all routers eventually synchronize their topology databases
- ☐ This is only possible in distance vector protocols
- ☐ Convergence in link-state protocols requires manual reconfiguration

Feedback

Correct Answer! Well done.



An engineer configures IPv6 address 2001:0db8:0000:0000:0000:ff00:0042:8329 *
for a server. Which of the following is the correctly compressed representation of
this address following IPv6 shorthand rules?

- ☐ 2001:db8::ff00:42:8329
- ☒ 2001:db8:0:0:ff00:42:8329
- ☐ 2001::db8:ff00:42:8329
- ☐ 2001:db8::ff00::42:8329

Feedback

Incorrect Answer. Please review and try again!

A host needs to determine whether another IP address, 172.16.45.200, is on its *
own local subnet ([172.16.45.0/23](#)) or must be reached via a router. Which
calculation should the host perform to decide this?

- ☐ Compare only the first octet of both addresses
- ☒ Perform a bitwise AND of both addresses with the subnet mask and compare the
resulting network portions
- ☐ Always assume the destination is remote
- ☐ Send an ARP broadcast without checking the subnet at all

Feedback

Correct Answer! Well done.



A network is designed so that if the primary link between two routers fails, Distance Vector routing takes 30-45 seconds to converge (due to periodic updates and timers), causing temporary packet loss, whereas Link State routing converges in a few seconds using triggered flooding. In a scenario demanding fast failover (e.g., financial trading network), which protocol family is the better analytical choice and why? *

- ☐ Distance Vector, because it uses less memory
- ☒ Link State, because triggered updates and independent shortest-path computation allow much faster convergence
- ☐ Either is equally suitable since convergence time doesn't matter
- ☐ Distance Vector, because periodic updates are more reliable

Feedback

Correct Answer! Well done.

An enterprise assigns [203.0.113.0/28](#) subnets to branch offices. One office requests support for exactly 14 hosts, no more, no less needed. Does a /28 subnet meet this requirement, and why? *

- ☐ No, /28 supports only 6 usable hosts
- ☒ Yes, /28 provides 16 total addresses, with 14 usable host addresses after reserving network and broadcast addresses
- ☐ No, /28 supports 30 usable hosts, wasting addresses
- ☐ Yes, but only if IPv6 is used instead

Feedback

Correct Answer! Well done.



A packet's IPv4 header shows a TTL of 64 when leaving the source and TTL of 60 * when it arrives at the destination. How many router hops did the packet traverse, and what mechanism made this observable?

- ☐ 6 hops; TCP sequence numbers
- ☒ 4 hops; each router along the path decrements the TTL field by 1
- ☐ 60 hops; ICMP redirect messages
- ☐ 4 hops; ARP resolution at each hop

Feedback

Correct Answer! Well done.

A university deploys IPv6 across campus and no longer needs NAT (Network Address Translation) for internet-facing devices, unlike their old IPv4 setup. What core reason explains why NAT becomes largely unnecessary with IPv6? *

- ☐ IPv6 disables all outbound internet traffic by default
- ☒ IPv6's vastly larger address space allows every device to have a unique global address, removing the address-scarcity reason NAT was originally used
- ☐ IPv6 addresses cannot be routed on the public internet
- ☐ NAT is a Layer 2 function unrelated to addressing scheme

Feedback

Correct Answer! Well done.



A distance-vector router receives conflicting hop-count information for the same destination network from two different neighbors within the same update interval. It selects the route with the lowest hop count and discards the other. Which fundamental distance-vector principle is being applied? *

- ☐ Longest prefix match
- ☒ Bellman-Ford-based selection of the minimum-cost (lowest metric) path among advertised routes
- ☐ Link-state flooding
- ☐ Administrative distance override

Feedback

Correct Answer! Well done.

A security team wants routers to automatically detect a specific type of network problem (e.g., unreachable host, redirect needed, or fragmentation needed) and report it back without requiring a full TCP/UDP session to be established. Which protocol operates at the network layer specifically to carry such diagnostic and error messages? *

- ☐ SMTP
- ☒ ICMP
- ☐ HTTP
- ☐ SNMP

Feedback

Correct Answer! Well done.



An organization has a Class B network [172.16.0.0/16](#) and wants to create subnets that each support up to 500 hosts, maximizing the number of subnets possible under that constraint. What subnet mask should be chosen, and roughly how many subnets result? *

- ☒ /23, yielding 128 subnets with 510 usable hosts each
- ☐ /24, yielding 256 subnets with 254 usable hosts each (too few hosts)
- ☐ /22, yielding 64 subnets with 1022 usable hosts each
- ☐ /25, yielding 512 subnets with 126 usable hosts each (too few hosts)

Feedback

Correct Answer! Well done.

A router in a multicast-enabled network uses IGMP to learn that three hosts on its local subnet want to join a video-streaming multicast group, even though the router itself has no direct interest in that content. What is the router's role in this scenario? *

- ☐ The router blocks all multicast traffic by default for security
- ☒ The router tracks group membership via IGMP and forwards multicast traffic destined for that group onto the subnet where interested hosts reside
- ☐ The router converts the multicast traffic into unicast copies for each host at Layer 2
- ☐ The router ignores IGMP messages since multicast is handled only by hosts

Feedback

Correct Answer! Well done.



A network engineer must choose between deploying RIP (Distance Vector) and OSPF (Link State) for a large enterprise network with 200 routers and frequent topology changes. Analytically, why would OSPF scale better in this scenario? *

- ☐ OSPF has a hop-count limit of 15 like RIP, so both scale equally
- ☒ OSPF hierarchically organizes the network into areas and computes shortest paths locally with Dijkstra, avoiding RIP's slow convergence and 15-hop limitation
- ☐ RIP uses less bandwidth for updates than OSPF in large networks
- ☐ OSPF cannot support more than 15 routers, making RIP the better choice

Feedback

Correct Answer! Well done.

A host is assigned IPv6 address fe80::1a2b:3c4d:5e6f:7a8b. Without consulting any external documentation, a network engineer immediately recognizes this address's scope from its prefix. What type of address is this? *

- ☐ A global unicast address routable on the public internet
- ☒ A link-local address, valid only on the local network segment and not routed beyond it
- ☐ A multicast address for a specific group
- ☐ A loopback address equivalent to IPv4's 127.0.0.1

Feedback

Correct Answer! Well done.

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